

# Brian Lee

✉ Email: lee.brian.j9@gmail.com

📱 Mobile: (571) 279-7134

Website: leebri2n.github.io

## EXPERIENCE

---

### Research Data Analyst (Applied AI and Data Assurance)

Virginia Tech National Security Institute

Arlington, Virginia

Aug. 2023 - Aug. 2025

- Led development of **CODEX**, a Python package for machine learning data assurance, to public release and implemented core features that supported analyses in 4 peer-reviewed publications.
- Delivered successful demos of CODEX at two workshops and presented a publication on its application to measuring data coverage bias in facial recognition datasets at the 2025 IEEE Conference on Artificial Intelligence (CAI 2025).
- Designed a computer vision pipeline enabling an AI use case integrating YOLO (You Only Look Once) models into a kill web — a data structure modeling systems and components — constructed by **MIMIK**, a Python package for creating and simulating custom kill webs.
- Reduced **2+ hours** of programming work per user by creating three internal computer vision libraries with pipelines that each preprocess a different image dataset for compatibility with YOLO models.

### Research Intern

Virginia Tech National Security Institute

Arlington, Virginia

May - Aug. 2022

- Constructed an image cleaning pipeline used to scrape, filter, and preprocess **50,000+** real images from Instagram into datasets organized by a hashtag provided as input.
- Enhanced StyleGAN3 image generation quality (**22% reduction in average FID score**) by applying transfer learning when training on limited-size Instagram datasets for **75+ models**.

## EDUCATION

---

### Virginia Tech

**B.S. in Computational Modeling and Data Analytics**

Minors in Computer Science and Mathematics

Blacksburg, Virginia

Aug. 2019 - May 2023

GPA: **3.485**

## RELEVANT COURSEWORK

---

### Mathematics & Modeling

- Dynamical Systems, Numerical Analysis and Optimization, Fourier Series, PDEs, Modeling and Simulation of Biological Systems

## PROJECTS

---

### Understanding the Dynamics of Multiple Prey Species on a Predator-Prey System

Apr. - May 2023

- Implemented a system of difference equations for a population model with terms defining population increase rate and seasonal birthrate.
- Proved the existence of nontrivial equilibria for a predator-free ecosystem and for a coexisting ecosystem of both prey and predator species.
- Ran **25+** numerical simulations on the model in MATLAB to identify cases where prey populations oscillate, overpopulate, or go extinct.

### Comparing Latent Representations of Protein Sequencing Models (CMDA Capstone)

Aug. - Dec. 2022

- Applied PCA model to map protein sequence model output datasets with up to **1,900** features into tractable, lower-dimensional spaces.
- Trained a collection of Gaussian Mixture Models to cluster protein sequence representations generated by three different protein sequencing models on datasets with up to **530,000** sequences and determined dissimilarity of model representations by computing average distance.

## PUBLICATIONS

---

- E. Lanus, **B. Lee**, D. Steberg, J. Chandrasekaran, and L. J. Freeman, "CODEX: Testing Machine Learning with the Coverage of Data Explorer Tool," in 2025 IEEE Conference on Artificial Intelligence Testing (AITest), 2025, to appear.
- S. Mhatre, **B. Lee**, E. Lanus, and L. J. Freeman, "Coverage Metrics for Detecting Bias in Facial Recognition Datasets," in 2025 IEEE Conference on Artificial Intelligence (CAI), May 2025.
- Lanus, E., **B. Lee**, Pol, L., Sobien, D., Kauffman, J., and Freeman, L.J. "Coverage for Identifying Critical Metadata in Machine Learning Operating Envelopes." 2024 IEEE International Conference on Software Testing, Verification, and Validation Workshops. Toronto, ON, Canada, 2024, pp. 217-226, doi: 10.1109/ICSTW64639.2025.10962464.
- **Lee, B.**, Eisenbeiser, L., and Lanus, E. "Training Generative Adversarial Networks on Small Datasets by way of Transfer Learning," *The ITEA Journal of Test and Evaluation*. 2023.